

ALİSTIRMALAR 2 - LIMIT

$$1. \lim_{n \rightarrow \infty} \frac{1-2+3-4+5-6+\dots-2n}{\sqrt{n^2+1} + \sqrt{4n^2-1}} = ?$$

$$\lim_{n \rightarrow \infty} \frac{1+3+5+\dots+2n-1 - (2+4+6+\dots+2n)}{\sqrt{n^2+1} + \sqrt{4n^2-1}}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2 - n(n+1)}{\sqrt{n^2+1} + \sqrt{4n^2-1}} = \lim_{n \rightarrow \infty} \frac{n^2 - n^2 - n}{\sqrt{n^2(1+\frac{1}{n^2})} + \sqrt{n^2(4-\frac{1}{n^2})}}$$

$$= \lim_{n \rightarrow \infty} \frac{-n}{\frac{1}{n}(\sqrt{1+\frac{1}{n^2}} + \sqrt{4-\frac{1}{n^2}})} = \lim_{n \rightarrow \infty} \frac{-1}{\sqrt{1+\frac{1}{n^2}} + \sqrt{4-\frac{1}{n^2}}} = \frac{-1}{1+2} = -\frac{1}{3}$$

$$2. \lim_{x \rightarrow 0} \frac{(1+x)^{1/3} - (1-x)^{1/3}}{x} = ?$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2) \Rightarrow (a-b) = \frac{a^3 - b^3}{(a^2 + ab + b^2)}$$

$$(1+x)^{1/3} - (1-x)^{1/3} = \frac{[(1+x)^{1/3}]^3 - [(1-x)^{1/3}]^3}{(1+x)^{2/3} + (1+x)^{1/3}(1-x)^{1/3} + (1-x)^{2/3}}$$

$$\lim_{x \rightarrow 0} \frac{(1+x)^{1/3} - (1-x)^{1/3}}{x} = \lim_{x \rightarrow 0} \frac{(1+x) - (1-x)}{x \left[(1+x)^{2/3} + (1+x)^{1/3}(1-x)^{1/3} + (1-x)^{2/3} \right]}$$

$$= \lim_{x \rightarrow 0} \frac{2x}{x \left[(1+x)^{2/3} + (1+x)^{1/3}(1-x)^{1/3} + (1-x)^{2/3} \right]} = \frac{2}{1+1+1} = \frac{2}{3}$$

$$3. \lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos 2x}}{\sqrt{2}x} = ?$$

$$\cos 2x = \cos^2 x - \sin^2 x \quad \sin^2 x + \cos^2 x = 1$$

$$\lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos^2 x + \sin^2 x}}{\sqrt{2}x} = \lim_{x \rightarrow 0} \frac{\sqrt{\sin^2 x + \sin^2 x}}{\sqrt{2}x}$$

$$= \lim_{x \rightarrow 0} \frac{\cancel{\sqrt{2}} |\sin x|}{\cancel{\sqrt{2}} x} \quad \nearrow \lim_{x \rightarrow 0^+} \frac{\sin x}{x} = 1$$

$$\searrow \lim_{x \rightarrow 0^-} \frac{|\sin x|}{x} = \lim_{h \rightarrow 0} \frac{|\sin(0-h)|}{(0-h)} = \lim_{h \rightarrow 0} \frac{|-\sin h|}{-h} = -1$$

$$\lim_{x \rightarrow 0^+} \frac{|\sin x|}{x} = 1$$

$\neq$ olduğundan limit mevcut değildir.

$$\lim_{x \rightarrow 0^-} \frac{|\sin x|}{x} = -1$$

$$4. \lim_{x \rightarrow -\infty} \left[\frac{x^4 \sin(1/x) + x^2}{1 + |x|^3} \right] = ?$$

$$\lim_{x \rightarrow -\infty} \frac{\cancel{x^3} \left[\frac{\sin \frac{1}{x}}{\frac{1}{x}} + \frac{1}{x} \right]}{\cancel{x^3} \left[\frac{1}{x^3} - 1 \right]} = \lim_{x \rightarrow -\infty} \frac{\overbrace{\sin \frac{1}{x}}^1}{\frac{1}{x}} + \frac{\cancel{1/x}}{\frac{1}{x^3} - 1} = \frac{1}{-1} = -1$$

$$5. \lim_{x \rightarrow \infty} \left(\frac{3x^2 + 2x + 1}{x^2 + x + 2} \right)^{\frac{6x+1}{3x+2}} = ?$$

$$\lim_{x \rightarrow \infty} \left[\frac{x^2 \left(3 + \frac{2}{x} + \frac{1}{x^2} \right)}{x^2 \left(1 + \frac{1}{x} + \frac{2}{x^2} \right)} \right]^{\frac{x(6 + \frac{1}{x})}{x(3 + \frac{2}{x})}}$$

$$= \left(\frac{3}{1} \right)^{\frac{6}{3}} = 3^2 = 9$$

$$6. \lim_{x \rightarrow \infty} \left[\sqrt{x + \sqrt{x + \sqrt{x}}} - \sqrt{x} \right] = ?$$

$$\lim_{x \rightarrow \infty} \left(\sqrt{x + \sqrt{x + \sqrt{x}}} - \sqrt{x} \right) \cdot \frac{(\sqrt{x + \sqrt{x + \sqrt{x}}} + \sqrt{x})}{(\sqrt{x + \sqrt{x + \sqrt{x}}} + \sqrt{x})}$$

$$= \lim_{x \rightarrow \infty} \frac{\cancel{x} + \sqrt{x + \sqrt{x}} - \cancel{x}}{(\sqrt{x + \sqrt{x + \sqrt{x}}} + \sqrt{x})} = \lim_{x \rightarrow \infty} \frac{\cancel{\sqrt{x}} \sqrt{1 + \frac{1}{\sqrt{x}}} \rightarrow 0}{\cancel{\sqrt{x}} \left(\sqrt{1 + \frac{1}{\sqrt{x}}} \sqrt{1 + \frac{1}{\sqrt{x}}} + 1 \right)} = \frac{1}{1+1} = \frac{1}{2}$$

$$7. f(x) = 4x - 5 \text{ fonksiyonu için}$$

eğer $0 < |x - 3| < \delta$ ise $|f(x) - 7| < \epsilon$ koşulunu sağlayan

$\delta > 0$ sayısını $\epsilon > 0$ sayısına bağlı olarak bulunuz.

$$|4x - 5 - 7| = |4x - 12| = |4(x - 3)|$$

$$= 4|x - 3| < \epsilon$$

$$|x - 3| < \delta$$

$$\Rightarrow |x - 3| < \frac{\epsilon}{4} = \delta$$

$$\Rightarrow \boxed{\delta = \frac{\epsilon}{4}} \text{ seçilebilir.}$$

$$8. \lim_{x \rightarrow \infty} \frac{x + \cos x}{x + \sin x} = ?$$

$$\lim_{x \rightarrow \infty} \frac{\cancel{x} \left(1 + \frac{\cos x}{x} \right)}{\cancel{x} \left(1 + \frac{\sin x}{x} \right)} = \lim_{x \rightarrow \infty} \frac{1 + \frac{\cos x}{x}}{1 + \frac{\sin x}{x}}$$

$$\left(\begin{array}{l} -1 \leq \cos x \leq 1 \\ -1 \leq \sin x \leq 1 \end{array} \right)$$

$$= \frac{1+0}{1+0} = 1$$

$$9. \lim_{x \rightarrow 2} \frac{2^x + 2^{3-x} - 6}{2^{\frac{x}{2}} - 2^{1-x}} = ?$$

$$\lim_{x \rightarrow 2} \frac{2^x + 2^3 \cdot 2^{-x} - 6}{\frac{1}{2^{\frac{x}{2}}} - \frac{2}{2^x}} = \lim_{x \rightarrow 2} \frac{2^x + 2^3 \cdot 2^{-x} - 6}{\frac{1}{2^x} \cdot (2^{\frac{x}{2}} - 2)}$$

$$= \lim_{x \rightarrow 2} \frac{2^{2x} - 6 \cdot 2^x + 8}{2^{\frac{x}{2}} - 2} \quad \left(\begin{array}{l} 2^x = t \\ x \rightarrow 2 \Rightarrow t \rightarrow 4 \end{array} \right)$$

$$= \lim_{t \rightarrow 4} \frac{t^2 - 6t + 8}{\sqrt{t} - 2} = \lim_{t \rightarrow 4} \frac{(t-4)(t-2)}{\sqrt{t} - 2} = \lim_{t \rightarrow 4} \frac{(\cancel{\sqrt{t}-2})(\sqrt{t}+2)(t-2)}{(\cancel{\sqrt{t}-2})}$$

$$= 4 \cdot 2 = 8$$

$$10. 0 < k < 1 \text{ olmak üzere } \lim_{n \rightarrow \infty} \frac{n^k \sin^2(n!)}{n+2} = ?$$

$$\lim_{n \rightarrow \infty} \frac{n^k \sin^2(n!)}{n(1 + \frac{2}{n})} = \lim_{n \rightarrow \infty} \frac{\overbrace{\sin^2(n!)}^{\text{sonlu bir sayı } (0 \leq A \leq 1)}}{\underbrace{n^{1-k}}_{\infty} \left(1 + \frac{2}{n}\right)} = 0$$

$$\left[\begin{array}{l} 0 \leq \sin^2(n!) \leq 1 \\ 1-k > 0 \\ n \rightarrow \infty \Rightarrow n^{1-k} \rightarrow \infty \end{array} \right]$$

$$11. \lim_{x \rightarrow \infty} \frac{2\sqrt{x} + 3\sqrt[3]{x} + 5\sqrt[5]{x}}{\sqrt{3x-2} + \sqrt[3]{2x-3}} = ?$$

$$\lim_{x \rightarrow \infty} \frac{\cancel{\sqrt{x}} (2 + 3 \cdot x^{-1/6} + 5 \cdot x^{-3/10})}{\cancel{\sqrt{x}} \left(\sqrt{3 - \frac{2}{x}} + x^{-1/6} \cdot \sqrt[3]{2 - \frac{3}{x}} \right)}$$

$$\lim_{x \rightarrow \infty} \frac{2 + \frac{3}{x^{1/6}} + \frac{5}{x^{3/10}}}{\sqrt{3 - \frac{2}{x}} + \frac{1}{x^{1/6}} \cdot \sqrt[3]{2 - \frac{3}{x}}} = \frac{2}{\sqrt{3}}$$

$$12. \lim_{n \rightarrow \infty} \frac{n!}{(n+1)! - n!} = ?$$

$$\lim_{n \rightarrow \infty} \frac{n!}{n!(n+1) - n!} = \lim_{n \rightarrow \infty} \frac{\cancel{n!}}{\cancel{n!}(n+1-1)} = \lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

$$13. \lim_{n \rightarrow \infty} \frac{\sqrt[4]{n^5+2} - \sqrt[3]{n^2+1}}{\sqrt[5]{n^4+2} - \sqrt{n^3+1}} = ?$$

$$\lim_{n \rightarrow \infty} \frac{n^{\frac{5}{4}} \left(\sqrt[4]{1+\frac{2}{n^5}} \right) - n^{\frac{2}{3}} \left(\sqrt[3]{1+\frac{1}{n^2}} \right)}{n^{\frac{4}{5}} \left(\sqrt[5]{1+\frac{2}{n^4}} \right) - n^{\frac{3}{2}} \sqrt{1+\frac{1}{n^3}}} \quad (\text{Enyüksek dereceli terim } n^{3/2})$$

$$= \lim_{n \rightarrow \infty} \frac{\cancel{n^{3/2}} \left[\underbrace{n^{-1/6}}_0 \sqrt[4]{1+\frac{2}{\cancel{n^5}}} - \underbrace{n^{-5/6}}_0 \sqrt[3]{1+\frac{1}{\cancel{n^2}}} \right]}{\cancel{n^{3/2}} \left[\underbrace{n^{-7/10}}_0 \sqrt[5]{1+\frac{2}{\cancel{n^4}}} - \underbrace{1}_0 \sqrt{1+\frac{1}{\cancel{n^3}}} \right]} = \frac{0-0}{0-1} = 0$$

$$14. \lim_{x \rightarrow \infty} \frac{\sqrt{x^2+1} - \sqrt[3]{x^2+1}}{\sqrt[4]{x^4+1} - \sqrt{x^4-1}} = ?$$

$$\lim_{x \rightarrow \infty} \frac{x \sqrt{1+\frac{1}{x^2}} - x^{2/3} \sqrt[3]{1+\frac{1}{x^2}}}{x \sqrt[4]{1+\frac{1}{x^4}} - x^{4/5} \sqrt[5]{1+\frac{1}{x^4}}} = \frac{1-0}{1-0} = 1$$

$$15. \lim_{x \rightarrow 1} \frac{\sqrt{1 - \cos 2(x-1)}}{x-1} = ?$$

$$x-1=t \quad \lim_{t \rightarrow 0} \frac{\sqrt{1 - \cos 2t}}{t} = \lim_{t \rightarrow 0} \frac{\sqrt{1 - \cos^2 t + \sin^2 t}}{t} = \lim_{t \rightarrow 0} \frac{\sqrt{2} |\sin t|}{t}$$

$$\lim_{t \rightarrow 0} \frac{\sqrt{2} |\sin t|}{t} \rightarrow \begin{cases} \lim_{t \rightarrow 0^+} \frac{\sqrt{2} \sin t}{t} = \sqrt{2} \\ \lim_{t \rightarrow 0^-} \frac{-\sqrt{2} \sin t}{t} = -\sqrt{2} \end{cases}$$

$\neq$ olduğundan limit mevcut değildir.

$$16. \lim_{t \rightarrow 0} \frac{\sin^2 t}{(1+\cos t)t} = ?$$

$$\lim_{t \rightarrow 0} \frac{\sin^2 t}{t^2} \cdot \frac{t}{1+\cos t} = \lim_{t \rightarrow 0} \underbrace{\left(\frac{\sin t}{t}\right)^2}_1 \cdot \underbrace{\frac{t}{1+\cos t}}_0 = 1 \cdot 0 = 0$$

$$17. \lim_{x \rightarrow \infty} \frac{\sin x}{x} = ?$$

$$\text{Her } x \in \mathbb{R} \quad -1 \leq \sin x \leq 1$$

$$\text{Her } x > 0 \text{ için } -\frac{1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}$$

$$\lim_{x \rightarrow \infty} -\frac{1}{x} = 0, \quad \lim_{x \rightarrow \infty} \frac{1}{x} = 0 \quad \text{ve Sıkıştırma teoreminden,}$$

$$\lim_{x \rightarrow \infty} \frac{\sin x}{x} = 0 \quad \text{dır.}$$